DI725 Project Report

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***Abstract*—This project replaces the SigLIP vision transformer in PaliGemma with lightweight CNNs to improve inference speed. Using MobileNetV3-Large and EfficientNet-B0 with single and multiple token strategies, four models are evaluated on the RISCM dataset. While performance slightly lags behind the original PaliGemma, the CNN-based models—especially the MobileNet-Multiple variant—offer a strong trade-off between speed and output quality, making them suitable for resource-constrained applications.**

***Index Terms*—transformers, vision language models**

I. Introduction

In this project, the vision encoder of PaliGemma [3] is replaced with lightweight CNN models. To observe the effect of the number of image tokens passed to the language decoder, two different tokenization methods are employed. The resulting four architectures are trained with the language model frozen and are evaluated using the BLEU, METEOR, and ROUGE metrics.

A. Literature Survey

PaliGemma is an open-source model that uses a 400M SigLIP vision transformer to encode images and a 2B Gemma auto-regressive decoder-only language model to generate outputs based on the provided image and text encodings for various tasks. PaliGemma is a highly versatile model capable of handling different tasks, including image captioning, visual question answering, video captioning, and referring expression segmentation. The SigLIP (Sigmoid Loss for Language-Image Pretraining) [8] model is an efficiency-improved version of the CLIP [7] that applies sigmoid loss to each image-text pair individually, rather than using softmax normalization across all image-text pairs in a batch. This approach results in more efficient training and better scalability. SigLIP-So-400M [1] is a shape-optimized vision transformer model used in PaliGemma to encode images into a format suitable for Gemma. Laurençon et al. [5] suggest that using pretrained vision encoders is crucial for training VLMs. This indicates that vision encoders play a critical role in VLMs. In the PaliGemma research paper, several ablation studies were conducted. One of these studies questions whether using an image encoder is truly necessary. They conducted an experiment by removing the image encoder and trained the model in Fuyu-style [2]. Fuyu-style refers to directly projecting raw pixel values into the decoder instead of encoding images using a transformer model. After training the model, they concluded that although this decoder-only model is not very training efficient -since it cannot reuse vision components-, it can still perform well even without pretraining. Since the Fuyu-style model eliminates the additional transformer model, it can operate much faster and more efficiently. This shows that using a vision transformer for encoding is not strictly necessary to achieve successful results.

In a study by Mishra et al. [6], the ResNet101 model is used to feed image representations into a transformer model for an image captioning task. This model uses a cross-modality attention architecture; that is, the image encodings are fed directly into a multi-head attention layer, similar to the vanilla encoder-decoder architecture of transformers.

In another study [4], it is argued that encoder-based VLMs face several potential issues, such as sensitivity to different image resolutions and aspect ratios, and high computational requirements. They also mention that eliminating encoders entirely slows down convergence and results in performance issues. They claim this is primarily due to the lack of high quality image-text data needed to train such models universally, and the inability to provide the model with high level image representations. In their model, EVE, they use a patch embedding layer consisting of 2D convolution, average pooling, and cross-attention layers to construct high-level image representations in a computationally efficient way. These projected layers are then fed into the language model along with the text encodings.

B. Project Proposal

The literature survey highlights a trade-off in current vision-language models (VLMs): while vision transformer-based encoders like SigLIP offer rich image representations, they are computationally expensive and slow. On the other hand, models that discard image encoding entirely—such as Fuyu-style architectures—sacrifice training efficiency and generalization by losing reusable visual representations.

This project proposes an alternative image encoding strategy aimed at balancing computational efficiency with training effectiveness. Specifically, I propose replacing the vision transformer encoder in PaliGemma with a lightweight CNN-based encoder. This encoder will transform raw image pixels into intermediate representations suitable for input into the language model decoder. CNNs are well-known for their computational efficiency and inductive biases suited for spatial data, making them a promising alternative to transformer-based vision encoders, especially in domain-specific tasks like satellite image captioning.

1) Methodology:

- **Encoder Design:** Two different lightweight convolutional neural network (CNN) models will be selected to encode satellite images into feature representations. These models will replace the SigLIP model in the original PaliGemma architecture.
- **Integration with Language Model:** The CNN outputs will be projected into a format compatible with the decoder-only language model in PaliGemma. To assess the importance of the number of image tokens fed into the language decoder, two different tokenization strategies will be employed. One approach will use a single image token, while the other will use multiple image tokens extracted from the feature layers of the CNN models.
- **Training and Evaluation:** The models will be trained on the RISCM dataset using the existing caption annotations. The language model will remain frozen to accelerate training. Performance will be evaluated using captioning metrics such as BLEU, METEOR, and ROUGE, as well as computational metrics like inference speed.
- **Comparative Analysis:** The performances of the models with different image encoders and tokenization strategies will be compared against the performance of the pretrained PaliGemma model.

2) Novelty and Conceptual Merits: By introducing a middle ground between full transformer encoders and no encoders, this project explores a novel architecture that leverages the efficiency of CNNs without compromising training dynamics. Developing a lightweight model for the classification of satellite images can be beneficial in real-world scenarios with limited computational resources.

Also, this study investigates how varying the number of image tokens fed into the model affects its performance, a factor that has not been systematically explored in prior work to my knowledge.

Finally, this approach offers insights into how hybrid architectures can enhance the efficiency and adaptability of vision-language models for specialized domains.

II. Dataset

The dataset used in this project is the RISCM dataset, which contains 44,521 remote sensing images with a fixed resolution of 224×224 pixels, accompanied by a total of 222,605 captions, five for each image.

A. Preprocessing Dataset

After inspecting the dataset, it was observed that some images had duplicate captions, which could introduce bias toward those captions. Therefore, they were removed from the dataset. To identify low-quality descriptions, a pretrained CLIP [2] model was used to compute similarity scores between images and captions. The image-caption pairs with the lowest similarity scores were examined, and it was found that the captions accurately represented the images. As a result, all captions were retained.

B. Final Dataset

Despite removing duplicate captions during preprocessing, the implementation using the Hugging Face datasets library required including all captions. Therefore, the complete dataset was ultimately retained, including originally duplicated captions, for consistency and compatibility with the library.

III. Modeling

A. Architecture

In the codebase, a model named CNNGemma is implemented. This model supports different architectures and tokenization methods, which are specified through configuration. A separate JSON configuration file is created for each model variant and saved within the codebase. Based on the architecture defined in the configuration, CNNGemma loads the corresponding pretrained weights as the image encoder. Additionally, the model can adapt to the specified tokenization method. Pretrained weights include the decoder weights of pretrained PaliGemma [3], and encoder weights trained with ImageNet-1k.

The following approaches are used to obtain single and multiple tokens:

- **Single Token:** The final layer of the CNN encoder, just before the classifier, is extracted and used as a single image token.
- **Multiple Tokens:** The final feature layer of the CNN encoder is reshaped into 49 image tokens.

As a result, four different architectures are created as follows:

- CNNGemma-MobileNet-Single
- CNNGemma-MobileNet-Multiple
- CNNGemma-EfficientNet-Single
- CNNGemma-EfficientNet-Multiple

B. Training

The initial source code lacks functionality for training the model. Therefore, it was updated and made compatible with the Hugging Face Transformers library.

The models are trained for 1 epoch with different settings to tune the hyperparameters. After exploring several configurations, the following hyperparameters were selected for training:

- **Optimizer:** Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$
- **Batch Size:** 1
- **Gradient Accumulation Steps:** 4
- **Initial Learning Rate:** 0.0001
- **Mixed Precision:** bfloat16
- **Learning Rate Schedule:** A cosine decay schedule was used for the learning rate as recommended in PaliGemma paper.
- **Warmup Strategy:** Warmup ratio of 0.05, with 0 warmup steps

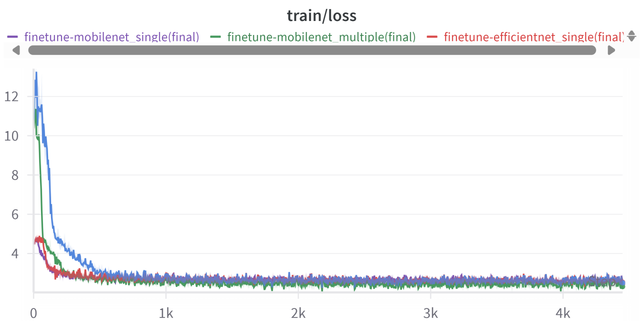

Fig. 1. Train Losses of Models

- **Weight Decay:** 0.0001

With the final hyperparameters, all the models are trained for 5 epochs with the training dataset sampled by randomly selecting one caption per image in each training step. This sampling strategy effectively results in a single full pass over the dataset (i.e., 1 effective epoch). The model weights are available on Hugging Face under the following model IDs: "ahmetbekcan/CNNGemma-MobileNet-Single-224", "ahmetbekcan/CNNGemma-MobileNet-Multiple-224", "ahmetbekcan/CNNGemma-EfficientNet-Single-224", and "ahmetbekcan/CNNGemma-EfficientNet-Multiple-224".

The training loss graphs can be seen in the Figure 1.

IV. Evaluation

BLEU, ROUGE, and METEOR scores for each model, including pretrained PaliGemma, are evaluated using the test split of the dataset. Each model's inference is compared against all available reference captions, and the minimum, maximum, and average scores are recorded. Also, inference speeds of different image encoders are evaluated. All results have been logged into Weights & Biases.

V. Results

A. Inference Speed

The inference speed of different image encoders was measured using dummy inputs with a batch size of 32. Each dummy input was passed through the image encoders 100 times, and throughput was calculated at each step. The comparison results can be seen in the Figure 2, and Table I

TABLE I
Throughput (FPS) Comparison of Image Encoders

Image Encoder	Throughput (FPS)
MobileNetV3-Large	3487
EfficientNet-B0	1584
SigLIP	219

Among the three encoders, MobileNetV3-Large was found to be the fastest, followed by EfficientNet-B0. Both models are significantly faster than SigLIP, demonstrating that replacing the image encoder with a lightweight alternative can substantially improve inference speed.

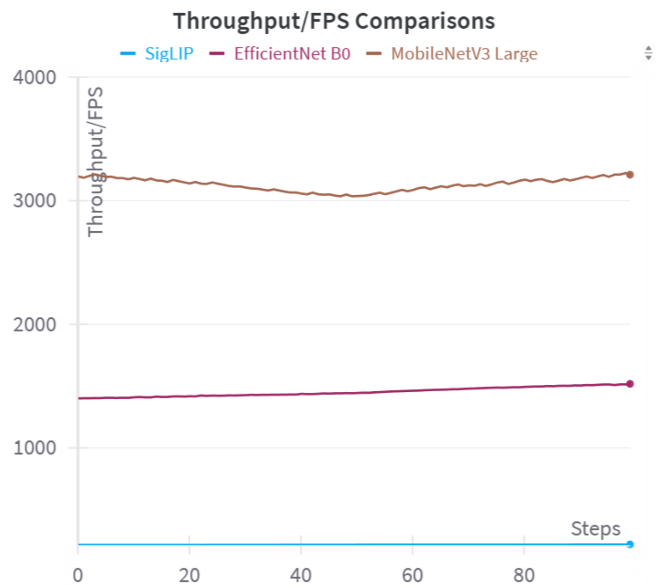

Fig. 2. Throughput comparisons of different image encoders

B. Model Performances

The inference results of different models can be seen in the Table II. While calculating the metrics, the maximum score across all references is used for each prediction.

Model	BLEU	ROUGE-L	METEOR	Average
MobileNet-Single-224	0.0004	0.2008	0.1269	0.1094
MobileNet-Multiple-224	0.0021	0.2206	0.1620	**0.1282**
EfficientNet-Single-224	0.0000	0.2076	0.1159	0.1078
EfficientNet-Multiple-224	0.0006	0.1720	0.1186	0.0971
PaliGemma (pretrained)	0.0045	0.2408	0.1890	**0.1448**

TABLE II
Comparison of Evaluation Metrics Across Different Model Configurations.

VI. Discussion

Although the fine-tuned models did not surpass the performance of the pre-trained PaliGemma, the difference is negligible considering that the CNN encoders were trained on a much smaller dataset than SigLIP.

The inference speed of CNN-based image encoders is significantly higher than that of SigLIP, making them a suitable choice for tasks where speed is critical.

Among the evaluated models, the MobileNet-Multiple variant demonstrates the best overall performance, effectively balancing inference speed and output quality. Notably, using multiple image tokens decreased the output quality of EfficientNet by approximately 11%, while increasing that of MobileNet by 17%. These results suggest that the effectiveness of tokenization strategies can vary depending on the underlying image encoder.

VII. Conclusion

This project investigated the feasibility of replacing the vision transformer encoder in PaliGemma with lightweight CNN-based alternatives. The results demonstrate that while these models do not outperform the original SigLIP-based architecture in terms of captioning metrics, they offer a compelling trade-off by significantly improving inference speed.

The findings show that CNN encoders, when combined with proper tokenization strategies, can serve as efficient and practical alternatives in vision-language models—especially in resource-constrained environments.

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